GAME BASED APTITUDE

60 PYQ-Style Questions • Practice Set 3

Placement & Competitive Exam Practice Series

Included	60 PYQ-Style Questions • Placement-focused
Useful For	TCS NQT • Cognizant • Infosys • Wipro • Accenture • Campus Placements
Topics Covered	Grid Challenge • Digit Challenge • Motion Challenge • Geo-Symmetry • Switch Challenge • Inductive Patterns
Format	10 MCQs + Answer Key per Page • Practice Set 3

These are PYQ-style practice questions modeled on common campus recruitment exam patterns.
They are designed for comprehensive practice, mental speed, and spatial logic assessment.

GAME BASED APTITUDE • GRID CHALLENGE & VISUAL MEMORY

Practice Set 3 • Section 1 of 6 • 10 Questions with Answer Key

Q1. In a 6x6 spatial game, how many total squares of all sizes (1x1, 2x2, ..., 6x6) exist?

A) 91 squares (sum of squares 1..6)
B) 36 squares
C) 72 squares
D) 84 squares

Q2. A token starts at center of a 5x5 grid (3, 3). Moves: 2 Up, 1 Left, 1 Down, 2 Right. Final position is:

A) (2, 4)
B) (1, 4)
C) (2, 2)
D) (4, 4)

Q3. In a dual-task memory game, user memorizes shape sequence while verifying math equations. This tests:

A) Working memory capacity (Central Executive)
B) Sensory receptor threshold
C) Muscle memory
D) Retinal persistence

Q4. Rotating a point (x, y) around the origin (0,0) by 180 degrees yields coordinates:

A) (-x, -y)
B) (y, -x)
C) (-y, x)
D) (x, -y)

Q5. In a pattern matching round, two identical shapes are presented with one rotated. Response time increases linearly with:

A) Angular rotation difference (Mental Rotation phenomenon)
B) Shape color saturation
C) Background grid line weight
D) Screen refresh rate

Q6. A 4x4 matrix contains binary digits 0 and 1. If each row has an even number of 1s, what parity check is used?

A) Even Parity
B) Odd Parity
C) Hamming code
D) Checksum 16

Q7. In an 8-tile sliding puzzle (3x3 grid with 1 blank), what is the goal configuration known as?

A) Solved state (1 through 8 in order)
B) Inversion state
C) Parity mismatch
D) Deadlock

Q8. A path on a grid moves strictly Right and Down from top-left (0,0) to bottom-right (3,3). Number of unique paths is:

A) 20 paths (6 choose 3)
B) 16 paths
C) 12 paths
D) 24 paths

Q9. In a strobe memory test, a visual target flashes for 50 milliseconds. Which eye movement rapidly redirects gaze?

A) Saccadic eye movement
B) Smooth pursuit
C) Nystagmus
D) Pupillary reflex

Q10. A symmetrical pattern in a 4x4 grid has 4 shaded cells. How many cells are in unshaded background?

A) 12 cells
B) 16 cells
C) 8 cells
D) 10 cells

ANSWER KEY & SOLUTIONS — GRID CHALLENGE & VISUAL MEMORY (SET 3)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • DIGIT CHALLENGE & MENTAL ARITHMETIC

Practice Set 3 • Section 2 of 6 • 10 Questions with Answer Key

Q1. Form target 50 using digits [5, 5, 2] with standard arithmetic:

A) (5 + 5) * 5 (no 2)
B) 5 * 5 * 2 = 50
C) 50 / 2 + 5
D) 5^2 + 5

Q2. Mental math speed: What is 48 * 25 calculated quickly by (48 / 4) * 100?

A) 1200
B) 1250
C) 1100
D) 1150

Q3. Find missing term: 100, 96, 88, 72, 40, ? (subtracting 4, 8, 16, 32, 64):

A) -24
B) 0
C) 8
D) -16

Q4. Evaluate: 125 + 237 + 75 + 63 = ?

A) 500
B) 490
C) 510
D) 480

Q5. If 2x - 7 = 33, what is x?

A) 20
B) 18
C) 22
D) 19

Q6. In a rapid fractional arithmetic game, what is (3/4) - (1/3)?

A) 5/12
B) 1/2
C) 2/7
D) 7/12

Q7. What is the average of numbers [14, 18, 22, 26, 30]?

A) 22 (middle number in symmetric AP)
B) 24
C) 20
D) 23

Q8. A store gives 20% discount on a $150 item, then applies 10% tax on discounted price. Final price is:

A) $132
B) $130
C) $135
D) $128

Q9. Evaluate mentally: Square root of 1764:

A) 42 (since 40^2=1600, ends in 4)
B) 44
C) 38
D) 48

Q10. If x * y = 72 and x + y = 17 with x > y, then x - y equals:

A) 1 (9 and 8: 9 - 8 = 1)
B) 2
C) 3
D) 4

ANSWER KEY & SOLUTIONS — DIGIT CHALLENGE & MENTAL ARITHMETIC (SET 3)

Q1: B • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • MOTION CHALLENGE & TRAJECTORY PHYSICS

Practice Set 3 • Section 3 of 6 • 10 Questions with Answer Key

Q1. A train 150m long traveling at 54 km/h (15 m/s) crosses a stationary pole in:

A) 10 seconds
B) 12 seconds
C) 8 seconds
D) 15 seconds

Q2. Gear A (12 teeth) meshes with Gear B (24 teeth), which meshes with Gear C (36 teeth). Speed ratio A:C is:

A) 3 : 1
B) 2 : 1
C) 1 : 3
D) 4 : 1

Q3. In an obstacle avoidance game, minimum stopping distance of a vehicle quadruples when speed is:

A) Doubled (Kinetic energy proportional to v^2)
B) Tripled
C) Halved
D) Quadrupled

Q4. A ball rolls down an inclined plane with uniform acceleration. Distance traveled is proportional to:

A) Square of time (t^2)
B) Time (t)
C) Cube of time (t^3)
D) Square root of time

Q5. In a simulated vacuum, a 10kg bowling ball and a 0.01kg feather dropped from height h reach ground:

A) At the exact same time
B) Ball lands first
C) Feather lands first
D) Neither lands

Q6. A swimmer can swim at 2 m/s in still water. If swimming upstream against a 1 m/s current, effective velocity is:

A) 1 m/s
B) 3 m/s
C) 2 m/s
D) 0.5 m/s

Q7. In a rotating space station, simulated artificial gravity is produced by which inertial force?

A) Centrifugal effect
B) Gravitational lensing
C) Magnetic attraction
D) Friction

Q8. What happens to the angular velocity of a spinning ice skater who pulls their arms in tightly?

A) Increases (Conservation of Angular Momentum)
B) Decreases
C) Remains unchanged
D) Stops spinning

Q9. A billiard ball hits a cushion rail head-on (90 deg angle). After elastic rebound, direction is:

A) 180 degrees opposite
B) 90 degrees right
C) 45 degrees
D) Unchanged forward

Q10. If velocity of an object is constant in both magnitude and direction, its net force is:

A) Zero (Newton First Law)
B) Constant non-zero
C) Increasing
D) Infinite

ANSWER KEY & SOLUTIONS — MOTION CHALLENGE & TRAJECTORY PHYSICS (SET 3)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • GEO-SYMMETRY & SPATIAL FOLD REASONING

Practice Set 3 • Section 4 of 6 • 10 Questions with Answer Key

Q1. A rectangular sheet is rolled along its length into an open cylinder. What part of the rectangle becomes cylinder height?

A) The width of the rectangle
B) The diagonal
C) The length
D) Half the area

Q2. How many total faces does a triangular pyramid (tetrahedron) have?

A) 4 faces
B) 6 faces
C) 5 faces
D) 8 faces

Q3. If an analog clock shows 3:00, what time is shown when looking at its image in a vertical plane mirror?

A) 9:00
B) 3:00
C) 6:00
D) 12:00

Q4. A solid rectangular cuboid has dimensions 1x2x3. What is its total exterior surface area?

A) 22 units squared (2*(2+3+6))
B) 18 units squared
C) 24 units squared
D) 20 units squared

Q5. Which 2D geometric shape has infinite lines of reflective symmetry?

A) Circle
B) Regular Dodecagon
C) Square
D) Oval / Ellipse

Q6. Folding a sheet with two diagonal crease lines crossing at center divides the square into:

A) 4 congruent isosceles right triangles
B) 4 rectangles
C) 2 trapezoids
D) 8 triangles

Q7. A cylinder viewed orthographically from its circular side base appears as a:

A) Circle
B) Rectangle
C) Triangle
D) Oval

Q8. A cube has 12 edges. If each edge length is doubled from 2cm to 4cm, volume increases by a factor of:

A) 8x (2 cubed)
B) 4x
C) 2x
D) 16x

Q9. How many planes of symmetry does a uniform solid sphere possess?

A) Infinite planes
B) 12 planes
C) 6 planes
D) 3 planes

Q10. Which shape can be formed by slicing a cone with a plane parallel to its base?

A) Circle
B) Parabola
C) Hyperbola
D) Triangle

ANSWER KEY & SOLUTIONS — GEO-SYMMETRY & SPATIAL FOLD REASONING (SET 3)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • SWITCH CHALLENGE & LOGIC GATE NETWORKS

Practice Set 3 • Section 5 of 6 • 10 Questions with Answer Key

Q1. If inputs to an AND gate are A=1, B=1, and C=0, the 3-input AND gate output is:

A) 0
B) 1
C) High Impedance
D) Undefined

Q2. In a 2-way multiplexer (MUX), which input determines which data line is routed to output?

A) Select line (S)
B) Enable line only
C) Ground line
D) Power rail

Q3. What is the inverse of the NOR gate?

A) OR gate
B) AND gate
C) NAND gate
D) XOR gate

Q4. In digital electronics, what binary logic state does +5V or Vcc typically represent?

A) Logic 1 (HIGH)
B) Logic 0 (LOW)
C) Negative logic
D) Floating

Q5. Simplify the Boolean expression: A OR (A AND B):

A) A (Absorption Law)
B) B
C) A AND B
D) 1

Q6. A parity generator circuit using XOR gates outputs 1 whenever the number of 1s in input is:

A) Odd
B) Even
C) Zero
D) Greater than 2

Q7. A circuit has 3 switches in parallel. The circuit will conduct current as long as:

A) At least one switch is closed
B) All 3 switches are closed
C) Exactly 2 switches are closed
D) All switches are open

Q8. What is the output of an inverter whose input is connected to the output of an XNOR gate?

A) XOR gate
B) AND gate
C) OR gate
D) NOR gate

Q9. In a circuit diagnostic round, if output is 1 when input is 0, and output is 0 when input is 1, this component is:

A) NOT gate (Inverter)
B) Buffer
C) Diode
D) Capacitor

Q10. According to Boolean algebra idempotence law, A AND A evaluates to:

A) A
B) A^2
C) 1
D) 0

ANSWER KEY & SOLUTIONS — SWITCH CHALLENGE & LOGIC GATE NETWORKS (SET 3)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A

GAME BASED APTITUDE • INDUCTIVE REASONING & SEQUENCE PATTERNS

Practice Set 3 • Section 6 of 6 • 10 Questions with Answer Key

Q1. Identify next term in series: 1, 8, 27, 64, 125, ?

A) 216 (6 cubed)
B) 243
C) 343
D) 256

Q2. Pattern of symbols: Up Arrow -> Right Arrow -> Down Arrow -> Left Arrow -> ?. Next direction is:

A) Up Arrow (90 deg clockwise cycle)
B) Down Arrow
C) Diagonal Arrow
D) Left Arrow

Q3. Alphabet series: AZ, BY, CX, DW, ?

A) EV (opposites in alphabet)
B) FU
C) EU
D) FV

Q4. Series: 50, 45, 41, 38, 36, ? (subtracting 5, 4, 3, 2, 1):

A) 35
B) 34
C) 33
D) 32

Q5. Which geometric transformation flips an image across a line producing a mirror reflection?

A) Reflection
B) Translation
C) Rotation
D) Dilation

Q6. In a logic matrix puzzle, each row adds dots: Row 1 has 2, Row 2 has 4, Row 3 has 6. Dot rule is:

A) +2 dots per row (Arithmetic progression)
B) Double dots
C) Random dots
D) Constant

Q7. Find odd one out: 64, 125, 216, 343, 500. Which is NOT a perfect cube of an integer?

A) 500
B) 64
C) 125
D) 343

Q8. Sequence: 2, 3, 5, 7, 11, 13, 17, ?. What is the next prime number?

A) 19
B) 21
C) 23
D) 25

Q9. A cipher shifts each letter forward by 3 (Caesar Cipher): 'CAT' becomes:

A) FDW
B) ECV
C) GDX
D) FBV

Q10. Complete the ratio analogy: Square is to 4 as Octagon is to:

A) 8 (number of vertices/sides)
B) 6
C) 10
D) 12

ANSWER KEY & SOLUTIONS — INDUCTIVE REASONING & SEQUENCE PATTERNS (SET 3)

Q1: A • Q2: A • Q3: A • Q4: A • Q5: A

Q6: A • Q7: A • Q8: A • Q9: A • Q10: A